

University of Dhaka



Assignment - 1

Course Title: Advanced Econometrics-II
Course Code: 408

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Question:

Using Google Finance, download the data on the Bangladeshi Taka to the United States Dollar exchange rate (hint: first use Google Sheets to retrieve the data) from 1 January 2011 to 1 January 2026.

Task 1: Import the data to STATA

Task 2: Draw a line graph with the exchange rate on the Y-axis and time on the X-axis.

Task 3: Create a new variable **Diff_Ex**. The variable should include the first difference values of the exchange rate.

Task 4: Suggest what univariate ARMA model **the change in the exchange** rate follows.

Write the report on all these tasks. Upload the report and the STATA code as a single PDF.

Solutions:

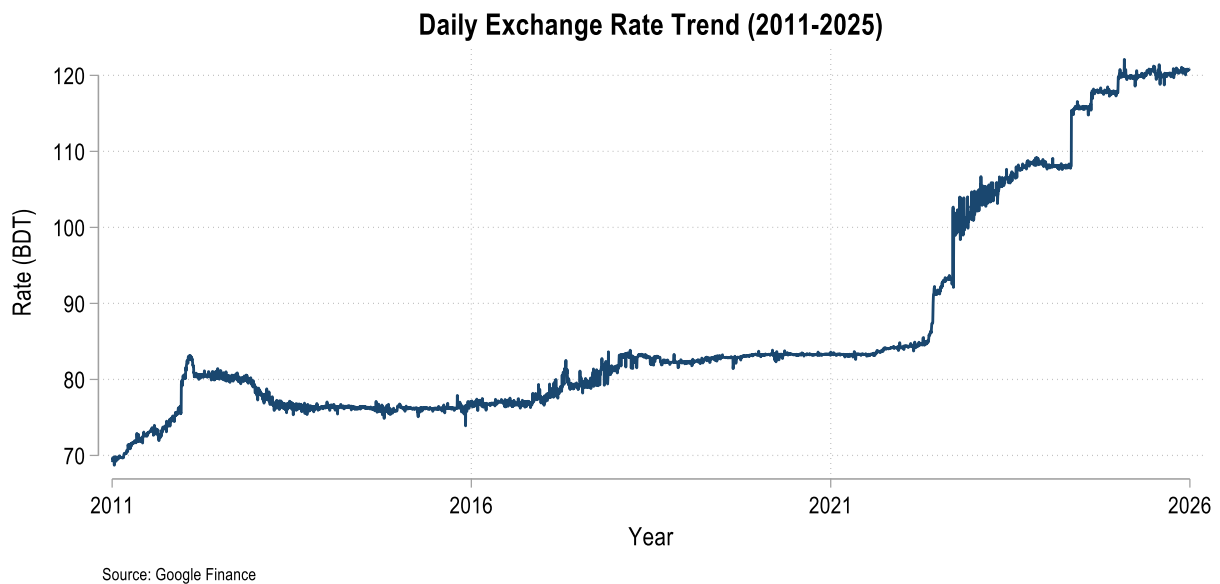
Task-1:

First, I have used Google finance to retrieve the USD to BDT exchange rate from 1 January 2011 to 1 January 2026, total 15 years of data. For this task I have used this formula,

```
=GOOGLEFINANCE("CURRENCY:USDBDT","price",DATE(2011,1,1),DATE(2026,1,1),"Daily")
```

There are 4480 daily observations starting from 3rd January 2011 to 31st December 2025. It excludes all the business holidays on Saturdays and Sundays. For calculation simplicity in Stata, I have converted the date format to an internal business calendar using Stata's *bcal* command. This will ensure that all the calculations acknowledge this time gap in date time column of the data.

Task-2:



At the start of the year 2011 the exchange rate was close to 70 BDT. Later it becomes 86 BDT and stayed unchanged with no trend for approximately 10 years. After the Covid-19 outbreak the exchange rate showed a huge upward trend from 86 BDT to 120 BDT and at the end of year 2025 it is consistent with the trend.

Task-3:

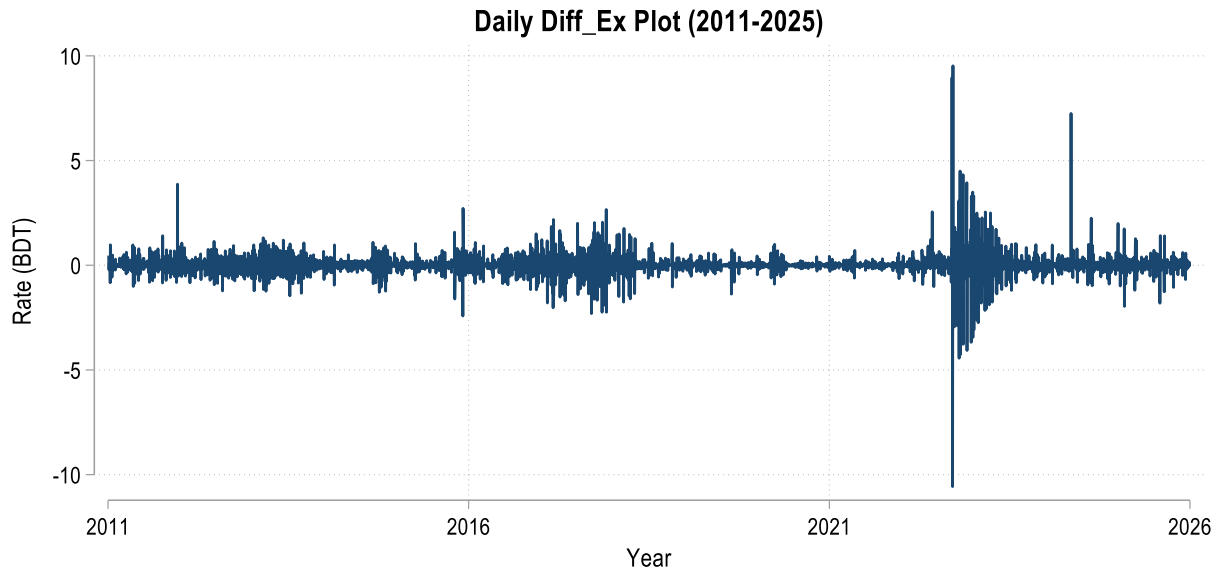
I used Stata's *generate Diff_Ex = D.Ex_Rate* command to create the first Difference of the Exchange Rate. Before that I converted the Date column into a business calendar to avoid the calculation miscreancy.

Task-4:

Ans: ARMA (3,2) Model.

Following the Box-Jenkins Methodology, I will follow (a) Identification (b) Estimation (c) Diagnostic, these 3 procedures to find the best Univariate ARMA model that the difference of the exchange rate follows.

(a) Identification

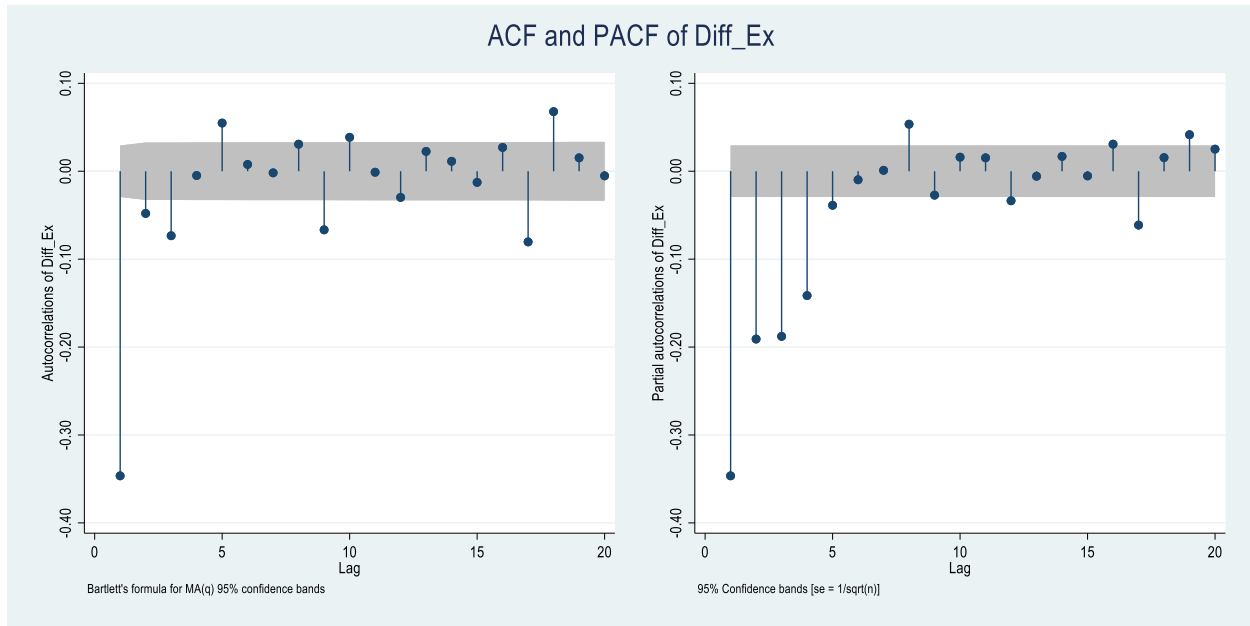


H0: Random walk without drift, $d = 0$

	Test	Dickey Fuller Critical Values		
	Statistics	1%	5%	10%
Z(t)	-84.872	-3.430	-2.860	-2.570

MacKinnon approximate p-value for $Z(t) = 0.0000$.

The Time Series plot of the first difference of exchange rate clearly shows no significant change in mean and no trend. And the Dicky-Fuller Test also rejects the Null Hypothesis of an Unit Root Process. So, the Diff_Ex series is stationary.



Both the Autocorrelation and Partial Auto-correlation Function of Diff_Ex variables show significant spikes till lag 3 then exponential decay. So, I assume the presence of ARMA (2,3), ARMA (3,2), ARMA (3,3), ARMA (3,4), ARMA (4,3). Now I will estimate all the assumed models and compare their information criteria to get the parsimonious model.

Information Criteria:

<i>Model</i>	<i>N</i>	<i>LL (model)</i>	<i>df</i>	<i>AIC</i>	<i>BIC</i>
ARMA (2,3)	4479	-3029.567	7	6073.134	6117.984
ARMA (3,2)	4479	-3028.1	7	6070.201	6115.051
ARMA (3,3)	4479	-3027.874	8	6071.749	6123.006
ARMA (3,4)	4479	-3027.815	9	6073.63	6131.295
ARMA (4,3)	4479	-3027.432	9	6072.865	6130.529

Among those guessed model , ARMA (3,2) yields the lowest AIC and BIC so this is my chosen Univariate model for the first difference of exchange rate.

(b) Estimation:

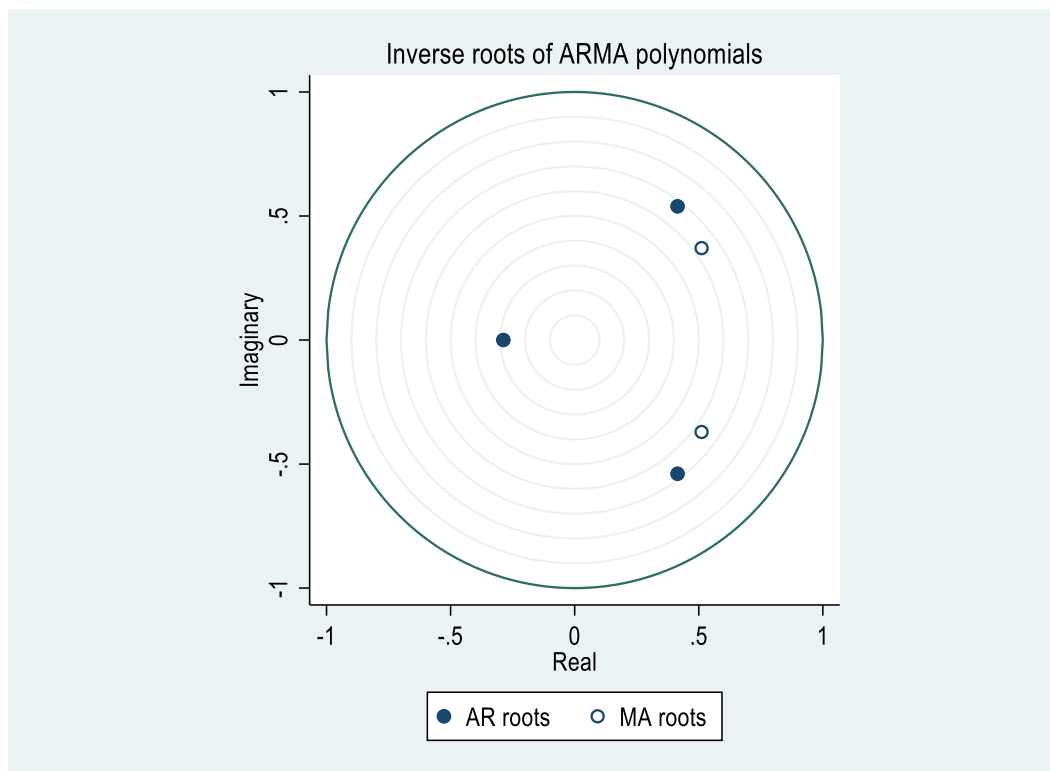
ARIMA (3,2) regression

Diff_Ex	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
Constant	.011	.004	2.89	.004	.004	.019	***
ar							
L	.542	.065	8.29	0	.414	.67	***
L2	-.224	.018	-12.33	0	-.26	-.188	***
L3	-.133	.011	-11.92	0	-.155	-.111	***
ma							
L	-1.023	.066	-15.49	0	-1.152	-.893	***
L2	.399	.046	8.58	0	.308	.49	***
Constant	.476	.001	465.0	0	.474	.478	***

2

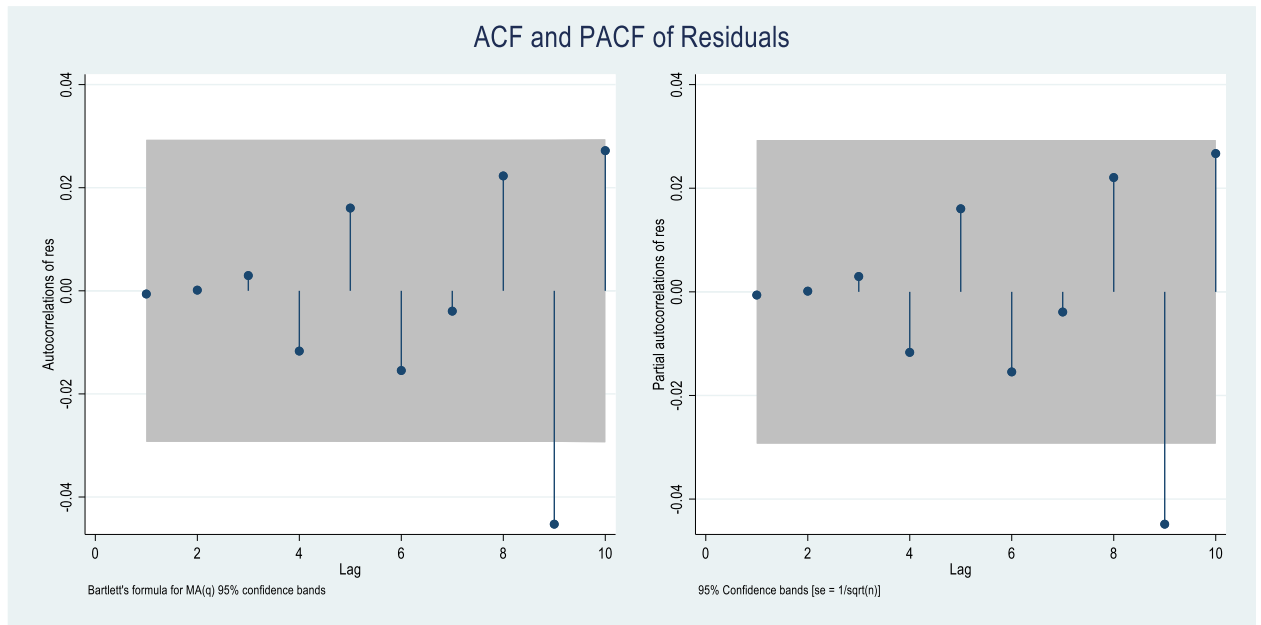
Mean dependent var	0.012	SD dependent var	0.532
Number of obs	4479	Chi-square	98684.671
Prob > chi2	.	Akaike crit. (AIC)	6070.201

*** $p < .01$, ** $p < .05$, * $p < .1$



After estimating the ARMA (3,2) model we can see all the lag coefficients of AR and MA process are statistically significant at 1% confidence intervals. And the roots diagram shows us that all the roots of AR and MA process are within the unit circle. So, we can say ARMA (3,2) model is acceptable for estimation.

(c) Diagnostic Test:



Portmanteau test for white noise

$$\text{Portmanteau (Q) statistics} = 17.7000$$

$$\text{Prob} > \text{chi2}(10) = 0.0602$$

The Ljung-Box test or Q test at lag 10 do not reject the Null Hypothesis of the residuals being white noise. The Autocorrelation and Partial Autocorrelation function of the residuals also justify this white noise process for lag 10. This diagnostic test satisfies the residuals being white noise and according to the steps of Box Jenkins Methodology ARMA (3,2) model is accepted as the univariate process of the difference of the exchange rate.

Ans: ARMA (3,2) model.

```

-----
> -----
      name: <unnamed>
      log: F:\ECON 408\Practice do file\Homework-1.log
      log type: text
      opened on: 14 May 2026, 15:03:41

.
. clear all

. set seed 1000

.
. * Task 1:
. *-----
.
.
. import excel "USD to BDT.xlsx", firstrow clear
(2 vars, 4,480 obs)

.
. rename Close Ex_Rate

. label variable Ex_Rate "Exchange Rate"

.
. replace Date = dofc(Date)
(4,480 real changes made)

. format Date %td

. tsset Date

Time variable: Date, 03jan2011 to 31dec2025, but with gaps
      Delta: 1 day

.
.
. * Task 2:
. *-----
. set scheme cleanplots

.
. tsline Ex_Rate, ///
>         lcolor(navy) lwidth(medthick) ///
>         title("{bf:Daily Exchange Rate Trend (2011-2025)}", size(large)) ///
>         xtitle("Year", size(medlarge)) ///
>         ytitle("Rate (BDT)", size(medlarge)) ///
>         ylabel(, labsize(medium)) ///
>         tlabel(01jan2011 01jan2016 01jan2021 01jan2026, format(%tdcY) labsize(med
> ium)) ///
>         graphregion(color(white)) xsize(8) ysize(4) ///
>         name(tsplot, replace) note("Source: Google Finance")

.
. graph export Hw_1_tsplot.emf, name(tsplot) replace
file F:\ECON 408\Practice do file\Hw_1_tsplot.emf saved as Enhanced Metafile format

.
.

```

```

. * Task 3:
. *-----
.
. bcal create bus_cal, from(Date) gen(time) replace
Business calendar bus_cal (format %tbbus_cal):

Purpose:

Range: 03jan2011 31dec2025
       18630      24106    in %td units
       0         4479    in %tbbus_cal units

Center: 03jan2011
       18630      in %td units
       0         in %tbbus_cal units

Omitted:      997      days
             66.5     approx. days/year

Included:     4,480     days
             298.8     approx. days/year

Notes:

Business calendar file bus_cal.stbcal saved.

Variable time created; it contains business dates in %tbbus_cal format.

. format time %tbbus_cal

. tsset time

Time variable: time, 03jan2011 to 31dec2025
Delta: 1 day

. gen Diff_Ex = D.Ex_Rate
(1 missing value generated)

.
.
. * Task 4:
. *-----
.
. * Step 1: Stationarity Check
.
. tsset Date

Time variable: Date, 03jan2011 to 31dec2025, but with gaps
Delta: 1 day

.
. set scheme cleanplots

.
. tsline Diff_Ex, ///
>     lcolor(navy) lwidth(medthick) ///
>     title("{bf:Daily Diff_Ex Plot (2011-2025)}", size(large)) ///
>     xtitle("Year", size(medlarge)) ///
>     ytitle("Rate (BDT)", size(medlarge)) ///
>     ylabel(, labsize(medium)) ///
>     tlabel(01jan2011 01jan2016 01jan2021 01jan2026, format(%tdcY) labsize(med
> ium)) ///
>     graphregion(color(white)) xsize(8) ysize(4) ///
>     name(Diff_Ex_plot, replace)

```

```
.
. graph export Diff_Ex_plot.emf, name(Diff_Ex_plot) replace
file F:\ECON 408\Practice do file\Diff_Ex_plot.emf saved as Enhanced Metafile format
```

```
.
. dfuller Diff_Ex
```

```
Dickey-Fuller test for unit root          Number of obs = 3,833
Variable: Diff_Ex                        Number of lags = 0
```

```
H0: Random walk without drift, d = 0
```

	Test	Dickey-Fuller		
	statistic	critical value	1%	5%
Z(t)	-84.872	-3.430	-2.860	-2.570

```
MacKinnon approximate p-value for Z(t) = 0.0000.
```

```
. tsset time
```

```
Time variable: time, 03jan2011 to 31dec2025
Delta: 1 day
```

```
.
. * Step 2: Empirical ACF and PACF check
. set scheme s2color
```

```
.
. ac Diff_Ex, name(acf_plot,replace) lags(20)
```

```
. pac Diff_Ex, name(pacf_plot,replace) lags(20)
```

```
. graph combine acf_plot pacf_plot, title("ACF and PACF of Diff_Ex") xsize(8) ysize(4)
> name(acf_pacf_plot, replace)
```

```
.
. graph export hw_1_acf_pacf_plot.emf, name(acf_pacf_plot) replace
file F:\ECON 408\Practice do file\hw_1_acf_pacf_plot.emf saved as Enhanced Metafile fo
> rmat
```

```
.
. *ARMA(2,3) , ARMA(3,2) , ARMA(3,3) , ARMA(3,4), ARMA(4,3)
```

```
.
. qui arima Diff_Ex, ar(1/2) ma(1/3)
```

```
. estat ic
```

```
Akaike's information criterion and Bayesian information criterion
```

Model	N	ll(null)	ll(model)	df	AIC	BIC
.	4,479	.	-3029.567	7	6073.134	6117.984

```
Note: BIC uses N = number of observations. See [R] BIC note.
```

```
.
```

```
. qui arima Diff_Ex, ar(1/3) ma(1/2)
```

```
. estat ic
```

Akaike's information criterion and Bayesian information criterion

Model	N	ll(null)	ll(model)	df	AIC	BIC
.	4,479	.	-3028.1	7	6070.201	6115.051

Note: BIC uses N = number of observations. See [R] BIC note.

```
. qui arima Diff_Ex, ar(1/3) ma(1/3)
```

```
. estat ic
```

Akaike's information criterion and Bayesian information criterion

Model	N	ll(null)	ll(model)	df	AIC	BIC
.	4,479	.	-3027.874	8	6071.749	6123.006

Note: BIC uses N = number of observations. See [R] BIC note.

```
. qui arima Diff_Ex, ar(1/3) ma(1/4)
```

```
. estat ic
```

Akaike's information criterion and Bayesian information criterion

Model	N	ll(null)	ll(model)	df	AIC	BIC
.	4,479	.	-3027.815	9	6073.63	6131.295

Note: BIC uses N = number of observations. See [R] BIC note.

```
. qui arima Diff_Ex, ar(1/4) ma(1/3)
```

```
. estat ic
```

Akaike's information criterion and Bayesian information criterion

Model	N	ll(null)	ll(model)	df	AIC	BIC
.	4,479	.	-3027.432	9	6072.865	6130.529

Note: BIC uses N = number of observations. See [R] BIC note.

```
.  
. * ARMA(3,2) Model has the lowest AIC and BIC  
. * Step 3: Estimation
```

```

. arima Diff_Ex, ar(1/3) ma(1/2)

(setting optimization to BHHH)
Iteration 0: log likelihood = -3040.2677
Iteration 1: log likelihood = -3031.423
Iteration 2: log likelihood = -3031.0705
Iteration 3: log likelihood = -3030.8361
Iteration 4: log likelihood = -3030.624
(switching optimization to BFGS)
Iteration 5: log likelihood = -3030.4353
Iteration 6: log likelihood = -3028.2888
Iteration 7: log likelihood = -3028.2146
Iteration 8: log likelihood = -3028.186
Iteration 9: log likelihood = -3028.1531
Iteration 10: log likelihood = -3028.1209
Iteration 11: log likelihood = -3028.1159
Iteration 12: log likelihood = -3028.1059
Iteration 13: log likelihood = -3028.104
Iteration 14: log likelihood = -3028.1029
(switching optimization to BHHH)
Iteration 15: log likelihood = -3028.1018
Iteration 16: log likelihood = -3028.101
Iteration 17: log likelihood = -3028.1009
Iteration 18: log likelihood = -3028.1008
Iteration 19: log likelihood = -3028.1008
(switching optimization to BFGS)
Iteration 20: log likelihood = -3028.1007
Iteration 21: log likelihood = -3028.1005
Iteration 22: log likelihood = -3028.1004
Iteration 23: log likelihood = -3028.1003
Iteration 24: log likelihood = -3028.1003
Iteration 25: log likelihood = -3028.1003
Iteration 26: log likelihood = -3028.1003
Iteration 27: log likelihood = -3028.1003
Iteration 28: log likelihood = -3028.1003
Iteration 29: log likelihood = -3028.1003

```

ARIMA regression

```

Sample: 04jan2011 thru 31dec2025      Number of obs   =      4479
Log likelihood = -3028.1                Wald chi2(5)     =    98684.67
                                      Prob > chi2       =      0.0000

```

Diff_Ex		OPG		z	P> z	[95% conf. interval]	
	Coefficient	std. err.					
Diff_Ex							
_cons	.0114831	.0039755	2.89	0.004	.0036914	.0192749	
ARMA							
ar							
L1.	.5417597	.065313	8.29	0.000	.4137485	.6697709	
L2.	-.2241	.0181753	-12.33	0.000	-.2597229	-.188477	
L3.	-.1330928	.0111702	-11.92	0.000	-.1549859	-.1111997	
ma							
L1.	-1.022842	.066048	-15.49	0.000	-1.152293	-.8933901	
L2.	.3989004	.0464674	8.58	0.000	.3078259	.4899748	
/sigma	.4757236	.001023	465.02	0.000	.4737185	.4777287	

Note: The test of the variance against zero is one sided, and the two-sided confidence interval is truncated at zero.

```
. estat ic
```

Akaike's information criterion and Bayesian information criterion

Model	N	ll(null)	ll(model)	df	AIC	BIC
.	4,479	.	-3028.1	7	6070.201	6115.051

Note: BIC uses N = number of observations. See [R] BIC note.

```
. estat aroots , name(arootshw1, replace)
```

Eigenvalue stability condition

Eigenvalue	Modulus
.4147067 + .5391681i	.680209
.4147067 - .5391681i	.680209
-.2876538	.287654

All the eigenvalues lie inside the unit circle.
AR parameters satisfy stability condition.

Eigenvalue stability condition

Eigenvalue	Modulus
.5114209 + .3706063i	.631586
.5114209 - .3706063i	.631586

All the eigenvalues lie inside the unit circle.
MA parameters satisfy invertibility condition.

```
. graph export arootshw1.emf, name(arootshw1) replace
```

file F:\ECON 408\Practice do file\arootshw1.emf saved as Enhanced Metafile format

```
. predict res , residuals  
(1 missing value generated)
```

```
. ac res, name(acf_res,replace) lags(10)
```

```
. pac res, name(pacf_res,replace) lags(10)
```

```
. graph combine acf_res pacf_res, title("ACF and PACF of Residuals") xsize(8) ysize(4)  
> name(hw1_res_ac_pac_plot,replace)
```

```
. graph export hw1_res_ac_pac_plot.emf, name(hw1_res_ac_pac_plot) replace
```

file F:\ECON 408\Practice do file\hw1_res_ac_pac_plot.emf saved as Enhanced Metafile format
> ormat

```
.  
. wntestq res, lags(10)
```

Portmanteau test for white noise

Portmanteau (Q) statistic =	17.7000
Prob > chi2(10) =	0.0602

```
.  
. log close  
name: <unnamed>  
log: F:\ECON 408\Practice do file\Homework-1.log  
log type: text  
closed on: 14 May 2026, 15:04:33
```

```
> -----
```